

MH1101 Tutorial 8 (Week 9)

Solution

Reference: Sections 4.3, 4.4 (Lecture Notes)

1. Find the limit of the sequence:

$$\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots$$

Solution Let a_n be the n -th term in the sequence. We have

$$a_{n+1} = \sqrt{2a_n}, \quad \text{for all } n \geq 1,$$

where $a_1 = \sqrt{2}$.

Note that $a_1 < 2$. Assume that $a_n < 2$. Then $a_{n+1} = \sqrt{2a_n} < \sqrt{2 \cdot 2} = 2$. Thus, by induction, $a_n < 2$ for all n . On the other hand, clearly $a_n > 0$ for all n . So the sequence $\{a_n\}$ is bounded.

Next, we will show that the sequence is increasing. Clearly, $a_1 = \sqrt{2} < \sqrt{2\sqrt{2}} = a_2$. Assume that $a_{n-1} < a_n$ (induction hypothesis).

By induction hypothesis, we have $a_{n+1} = \sqrt{2a_n} > \sqrt{2a_{n-1}} = a_n$. Hence, the sequence is increasing by induction on $n \geq 1$.

By the Monotonic Sequence Theorem, the sequence $\{a_n\}$ has a limit, say L . Applying limit on both sides of the recurrence, we have

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sqrt{2a_n}$$

$$L = \sqrt{2L}$$

$$L^2 - 2L = 0$$

$$L(L - 2) = 0.$$

Since $L > 0$ (as the sequence is increasing), we deduce that $L = 2$.

Alternative solution.

$$\begin{aligned}
a_n &= 2^{\frac{1}{2}} a_{n-1}^{\frac{1}{2}} \\
&= 2^{\frac{1}{2}} (2^{\frac{1}{2}} a_{n-2}^{\frac{1}{2}})^{\frac{1}{2}} \\
&= 2^{\frac{1}{2} + \frac{1}{2^2}} a_{n-2}^{\frac{1}{2^2}} \\
&= \vdots \\
&= 2^{\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}} a_1^{\frac{1}{2^{n-1}}} \\
&= 2^{\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n}} \\
&= 2^{1 - \frac{1}{2^n}}
\end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 2^{1 - \frac{1}{2^n}} = 2^{\lim_{n \rightarrow \infty} 1 - \frac{1}{2^n}} = 2.$$

2. Show that the sequence defined by

$$a_1 = 1, \quad a_{n+1} = 3 - \frac{1}{a_n}$$

is increasing and $0 < a_n < 3$ for all n . Deduce that the sequence $\{a_n\}$ is convergent and find its limit.

Solution We prove by induction that $\{a_n\}$ is increasing and $0 < a_n < 3$.

Let P_n be the statement that $a_{n+1} > a_n$ and $0 < a_n < 3$.

Clearly, P_1 is true:

$$a_2 = 3 - \frac{1}{a_1} = 3 - \frac{1}{1} = 2 > 1 = a_1, \quad \text{and} \quad 0 < a_1 = 1 < 3.$$

Assume P_n is true, that is the following is true:

$$a_{n+1} > a_n, \quad \text{and} \quad 0 < a_n < 3. \tag{1}$$

We want to show that P_{n+1} is true. Note that

$$\begin{aligned}
a_{n+2} &= 3 - \frac{1}{a_{n+1}} \\
&> 3 - \frac{1}{a_n} \quad (\text{since } a_{n+1} > a_n \text{ by (1)}) \\
&= a_{n+1} \quad (\text{by definition of } a_{n+1}).
\end{aligned}$$

Also

- $a_{n+1} > 0$ (since $a_{n+1} > a_n > 0$ by (1))
- $a_{n+1} = 3 - \frac{1}{a_n} < 3$ (since $a_n > 0$ by (1))

We have proved that

$$a_{n+2} > a_{n+1}, \quad \text{and} \quad 0 < a_{n+1} < 3,$$

that is the statement P_{n+1} is true.

By induction P_n is true for all n . Thus,

$$a_1 < a_2 < a_3 < \cdots$$

is increasing and $0 < a_n < 3$ for all n .

By the Monotonic Sequence Theorem, the sequence $\{a_n\}$ converges. Let

$$\lim_{n \rightarrow \infty} a_n = L.$$

Then

$$\lim_{n \rightarrow \infty} a_{n+1} = L.$$

Applying limit to the recurrence, we have

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left(3 - \frac{1}{a_n} \right) = 3 - \frac{1}{\lim_{n \rightarrow \infty} a_n}$$

$$L = 3 - \frac{1}{L}$$

$$L^2 - 3L + 1 = 0$$

$$L = \frac{3 \pm \sqrt{5}}{2}.$$

But $L > 1$, since $a_n > a_{n-1} > \cdots > a_1 = 1$, and so L being the limit of a_n cannot be smaller than 1. The number $\frac{3-\sqrt{5}}{2} < 1$ cannot be the limit, so the only possibility for L is

$$L = \frac{3 + \sqrt{5}}{2}.$$

3. Show that the sequence defined by

$$a_1 = 2, \quad a_{n+1} = \frac{1}{3 - a_n}$$

satisfies $0 < a_n \leq 2$, and is decreasing. Deduce that the sequence $\{a_n\}$ is convergent and find its limit.

Solution

We prove by induction that $\{a_n\}$ is decreasing and $0 < a_n \leq 2$.

Let P_n be the statement that $a_{n+1} < a_n$ and $0 < a_n \leq 2$.

Clearly, P_1 is true:

$$a_2 = \frac{1}{3 - a_1} = \frac{1}{3 - 2} = 1 < 2 = a_1, \quad \text{and} \quad 0 < a_1 = 2 \leq 2.$$

Assume P_n is true, that is the following holds:

$$a_{n+1} < a_n, \quad \text{and} \quad 0 < a_n \leq 2. \tag{2}$$

$$\begin{aligned} a_{n+2} &= \frac{1}{3 - a_{n+1}} \\ &< \frac{1}{3 - a_n} \quad (\text{since } a_{n+1} < a_n \leq 2) \\ &= a_{n+1} \quad (\text{by definition of } a_{n+1}). \end{aligned}$$

Also

- $a_{n+1} > 0$ (since $a_{n+1} = \frac{1}{3 - a_n}$ (by definition) and $a_n \leq 2$ (by (2)))
- $a_{n+1} \leq 2$ (since $a_{n+1} < a_n \leq 2$ by (2))

Hence, we have proved that

$$a_{n+2} < a_{n+1}, \quad \text{and} \quad 0 < a_{n+1} \leq 2,$$

that is the statement P_{n+1} is true.

By induction P_n is true for all n . Thus,

$$a_1 > a_2 > a_3 > \cdots$$

is increasing and $0 < a_n \leq 2$ for all n .

By the Monotonic Sequence Theorem, the sequence $\{a_n\}$ converges. Let

$$\lim_{n \rightarrow \infty} a_n = L.$$

Then

$$\lim_{n \rightarrow \infty} a_{n+1} = L.$$

Applying limit to the recurrence, we have

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left(\frac{1}{3 - a_n} \right) = \frac{1}{3 - \lim_{n \rightarrow \infty} a_n}$$

$$L = \frac{1}{3 - L}$$

$$L^2 - 3L + 1 = 0.$$

$$L = \frac{3 \pm \sqrt{5}}{2}.$$

But $L \leq 2$, since all the terms a_n is at most 2, and so L being the limit of a_n cannot be greater than 2. The number $\frac{3+\sqrt{5}}{2} > 2$ cannot be the limit, so the only possibility for L is

$$L = \frac{3 - \sqrt{5}}{2}.$$

4. Define $e_n = \left(1 + \frac{1}{n}\right)^n$. Show that the sequence $\{e_n\}_{n=1}^{\infty}$ is increasing.

Hint: Use the Binomial Theorem $(1 + a)^n = \sum_{k=0}^n \binom{n}{k} a^k$.

Solution Using the Binomial Theorem,

$$\begin{aligned} e_n &= \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^n \frac{n!}{(n-k)!k!} \frac{1}{n^k} \\ &= \sum_{k=0}^n \frac{n(n-1)(n-2) \cdots (n-k+1)}{n^k} \frac{1}{k!} \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=0}^n \frac{n}{n} \frac{(n-1)}{n} \frac{(n-2)}{n} \cdots \frac{(n-k+1)}{n} \frac{1}{k!} \\
&= \sum_{k=0}^n \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \frac{1}{k!} \\
&= \sum_{k=0}^n w_{1,n} \cdots w_{k-1,n} \frac{1}{k!},
\end{aligned}$$

where $w_{i,n} = 1 - \frac{i}{n}$.

Notice that $w_{i,n+1} > w_{i,n}$ for all $i = 1, \dots, k-1$. Hence

$$w_{1,n+1} w_{2,n+1} \cdots w_{k-1,n+1} > w_{1,n} w_{2,n} \cdots w_{k-1,n}.$$

It follows that

$$\begin{aligned}
e_{n+1} &= \sum_{k=0}^{n+1} w_{1,n+1} \cdots w_{k-1,n+1} \frac{1}{k!} \\
&= \sum_{k=0}^n w_{1,n+1} \cdots w_{k-1,n+1} \frac{1}{k!} + w_{1,n+1} \cdots w_{n,n+1} \frac{1}{(n+1)!} \\
&> \sum_{k=0}^n w_{1,n+1} \cdots w_{k-1,n+1} \frac{1}{k!} \\
&> \sum_{k=0}^n w_{1,n} \cdots w_{k-1,n} \frac{1}{k!} = e_n.
\end{aligned}$$

Hence, the sequence $\{e_n\}$ is increasing.

5. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$(i) \sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$$

$$(iv) \sum_{n=2}^{\infty} \left(\frac{1}{e^n} + \frac{2}{n^2 - 1} \right)$$

$$(ii) \sum_{n=1}^{\infty} \frac{6 \cdot 2^{2n-1}}{3^n}$$

$$(v) \sum_{n=1}^{\infty} \frac{n^3 + n^2}{n^3 - 2n + 5}$$

$$(iii) \sum_{n=1}^{\infty} \frac{e^n}{n^2}$$

$$(vi) \sum_{n=1}^{\infty} \left(\frac{3}{5^n} + \frac{2}{n} \right)$$

Solution

(i) Write

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n} = \frac{1}{4} \sum_{n=1}^{\infty} \left(-\frac{3}{4}\right)^{n-1}.$$

The series $\sum_{n=1}^{\infty} \left(-\frac{3}{4}\right)^{n-1}$ is a geometric series with the first term $a = 1$, and ratio $r = -\frac{3}{4}$. So it is convergent:

$$\sum_{n=1}^{\infty} \left(-\frac{3}{4}\right)^{n-1} = \frac{a}{1-r} = \frac{1}{1-(-3/4)} = \frac{4}{7}.$$

Thus,

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n} = \frac{1}{4} \sum_{n=1}^{\infty} \left(-\frac{3}{4}\right)^{n-1} = \frac{1}{4} \cdot \frac{4}{7} = \frac{1}{7}.$$

(ii) Write

$$\sum_{n=1}^{\infty} \frac{6 \cdot 2^{2n-1}}{3^n} = \sum_{n=1}^{\infty} \frac{6(2^2)^{n-1} 2}{3^n} = 3 \cdot \sum_{n=1}^{\infty} \left(\frac{4}{3}\right)^{n-1}.$$

The series $\sum_{n=1}^{\infty} \left(\frac{4}{3}\right)^{n-1}$ is a geometric series with ratio $r = \frac{4}{3} > 1$. So it diverges. Hence, the given series is also divergent.

(iii) Let $a_n = \frac{e^n}{n^2}$. Note that

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{e^n}{n^2} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{x^2} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2x} \quad (\text{by L'Hopital's Rule}) \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2} \quad (\text{by L'Hopital's Rule}) \\ &= \infty \neq 0. \end{aligned}$$

By the n -term Test for Divergence, the series $\sum a_n$ diverges.

(iv) Note that

$$\sum_{n=2}^{\infty} \frac{1}{e^n} = \sum_{n=2}^{\infty} \left(\frac{1}{e}\right)^n$$

is a geometric series with the first term $a = \frac{1}{e^2}$ and ratio $-1 < \frac{1}{e} < 1$.

So the series $\sum_{n=2}^{\infty} \frac{1}{e^n}$ converges to $\frac{a}{1-r} = \frac{\frac{1}{e^2}}{1-\frac{1}{e}} = \frac{1}{e(e-1)}$.

On the other hand,

$$\frac{2}{n^2 - 1} = \frac{1}{n - 1} - \frac{1}{n + 1}.$$

So the series

$$\sum_{n=2}^{\infty} \frac{2}{n^2 - 1} = \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \cdots$$

Let

$$s_k = \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \cdots + \frac{1}{k-1} - \frac{1}{k+1}.$$

In particular,

$$\begin{aligned} s_2 &= \frac{1}{1} - \frac{1}{3}, \\ s_3 &= \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4}, \\ s_4 &= \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5}, \\ s_5 &= \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6}, \\ s_6 &= \frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \frac{1}{5} - \frac{1}{7}. \end{aligned}$$

This sum is a telescoping sum:

$$s_k = \frac{1}{1} + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1}.$$

Thus,

$$\sum_{n=2}^{\infty} \frac{2}{n^2 - 1} = \lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \left(\frac{1}{1} + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1} \right) = \frac{1}{1} + \frac{1}{2} = \frac{3}{2}.$$

Summing up, we have

$$\sum_{n=1}^{\infty} \left(\frac{1}{e^n} + \frac{2}{n^2 - 1} \right) = \left(\sum_{n=2}^{\infty} \frac{1}{e^n} \right) + \left(\sum_{n=2}^{\infty} \frac{2}{n^2 - 1} \right) = \frac{1}{e(e-1)} + \frac{3}{2}.$$

(v) Let $a_n = \frac{n^3+n^2}{n^3-2n+5}$. Then

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{n^3 + n^2}{n^3 - 2n + 5} \\ &= \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{1 - \frac{2}{n^2} + \frac{5}{n^3}} \\ &= 1 \neq 0. \end{aligned}$$

By the n -th term Test for Divergence, the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(vi) Let $a_n = b_n + c_n$, where $b_n = \frac{3}{5^n}$, $c_n = \frac{2}{n}$. We show that $\sum a_n$ is divergent by contradiction.

Suppose for a contradiction that $\sum a_n$ is convergent. The series $\sum b_n$ is convergent since it is a geometric series with ratio $\frac{1}{5} < 1$. This implies that

$$\sum c_n = \sum (a_n - b_n) = \sum a_n - \sum b_n$$

is also convergent.

However, the series $\sum c_n = \sum \frac{2}{n} = 2 \sum \frac{1}{n}$. Since $\sum \frac{1}{n}$ is a Harmonic series, it is divergent. This implies that $\sum c_n$ is divergent, contradicting the preceding paragraph.

Hence, our initial assumption that $\sum a_n$ is convergent is not true. So $\sum a_n$ is divergent.

Answer Keys. **2.** $\frac{3+\sqrt{5}}{2}$ **3.** $\frac{3-\sqrt{5}}{2}$ **5.** (i) $\frac{1}{7}$ (ii) divergent (iii) divergent
(iv) $\frac{1}{e(e-1)} + \frac{3}{2}$ (v) divergent (vi) divergent